

Textual Description

The Airport Management System database is designed to store and manage information about airports, airlines, flights, passengers, bookings, boarding passes, baggage, security checks, baggage checks, and booking changes.

Each **airline** can operate many flights. Each **flight** belongs to one airline and has a departure airport and an arrival airport. A flight can have many bookings.

Each **passenger** can have many bookings. A booking connects a passenger with a specific flight and stores information such as booking status, booking platform, and ticket price.

Each **booking** can have a boarding pass, baggage records, baggage checks, and booking changes. Passenger information is also connected to security checks and baggage checks.

Primary keys (**PK**) uniquely identify each record in a table, while foreign keys (**FK**) are used to create relationships between tables. The database mainly uses one-to-many (**1:N**) relationships and a one-to-one (**1:1**) relationship between Booking and Boarding Pass.

AIRPORT – stores information about airports.

AIRLINE – stores information about airlines.

FLIGHT – stores flight and schedule information.

PASSENGER – stores personal information about passengers.

BOOKING – stores passenger flight reservations.

BOARDING_PASS – stores boarding pass and seat information.

BAGGAGE – stores information about passenger baggage.

BAGGAGE_CHECK – stores baggage check results.

SECURITY_CHECK – stores passenger security check results.

BOOKING_CHANGE – stores information about changes made to bookings.

LEGEND

PK – Primary Key

FK – Foreign Key

1 – One

N – Many

1:N – One-to-Many relationship

1:1 – One-to-One relationship

AIRPORT		
PK	<u>airport_id</u>	INTEGER
	airport_name	VARCHAR(100)
	country	VARCHAR(100)
	state	VARCHAR(100)
	city	VARCHAR(100)
	created_at	TIMESTAMP
	updated_at	TIMESTAMP

AIRLINE		
PK	<u>airline_id</u>	INTEGER
	airline_code	VARCHAR(10)
	name	VARCHAR(100)
	country	VARCHAR(100)
	created_at	TIMESTAMP
	updated_at	TIMESTAMP

FLIGHT		
PK	<u>flight_id</u>	INTEGER
FK	airline_id	INTEGER
FK	departure_airport_id	INTEGER
FK	arrival_airport_id	INTEGER
	departing_gate	VARCHAR(20)
	arriving_gate	VARCHAR(20)
	scheduled_departure	TIMESTAMP
	scheduled_arrival	TIMESTAMP
	actual_departure	TIMESTAMP
	actual_arrival	TIMESTAMP
	created_at	TIMESTAMP
	updated_at	TIMESTAMP

PASSENGER		
PK	<u>passenger_id</u>	INTEGER
	first_name	VARCHAR(100)
	last_name	VARCHAR(100)
	gender	VARCHAR(20)
	date_of_birth	DATE
	country_of_citizenship	VARCHAR(100)
	country_of_residence	VARCHAR(100)
	passport_number	VARCHAR(50)
	created_at	TIMESTAMP
	updated_at	TIMESTAMP

SECURITY_CHECK		
PK	<u>security_check_id</u>	INTEGER
FK	passenger_id	INTEGER
	check_result	VARCHAR(50)
	created_at	TIMESTAMP
	updated_at	TIMESTAMP

BOOKING		
PK	<u>booking_id</u>	INTEGER
FK	flight_id	INTEGER
FK	passenger_id	INTEGER
	status	VARCHAR(50)
	booking_platform	VARCHAR(50)
	ticket_price	DECIMAL(10,2)
	created_at	TIMESTAMP
	updated_at	TIMESTAMP

BOARDING_PASS		
PK	<u>boarding_pass_id</u>	INTEGER
FK	booking_id	INTEGER
	seat	VARCHAR(10)
	boarding_time	TIMESTAMP
	created_at	TIMESTAMP
	updated_at	TIMESTAMP

BAGGAGE_CHECK		
PK	<u>baggage_check_id</u>	INTEGER
FK	booking_id	INTEGER
FK	passenger_id	INTEGER
	check_result	VARCHAR(50)
	created_at	TIMESTAMP
	updated_at	TIMESTAMP

BOOKING_CHANGE		
PK	<u>booking_change_id</u>	INTEGER
FK	booking_id	INTEGER
	change_description	VARCHAR(255)
	created_at	TIMESTAMP
	updated_at	TIMESTAMP

BAGGAGE		
PK	<u>baggage_id</u>	INTEGER
FK	booking_id	INTEGER
	weight_kg	DECIMAL(5,2)
	created_at	TIMESTAMP
	updated_at	TIMESTAMP

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